LESSON PLAN

AIRCRAFT ELECTRICAL EQUIPMENT, POWER GENERATION AND DISTRIBUTION (15)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	08
Total No. of Lesson Plans	35	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	The earth's atmosphere and its characteristics
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	The earth’s atmosphere and its characteristics. Familiarization with Atmospheric pressure, Atmospheric temperature, Standard atmosphere & Measurement of atmospheric pressure.
5. Specific Objectives	15.1.3.1 Trainee will be able to explain earth’s Atmospheric pressure and Atmospheric temperature.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of Standard Atmosphere data tables; Instructor supervision required for data verification.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the earth's atmosphere?
(b) How does atmospheric pressure change with altitude?
(c) Why is temperature variation important for aircraft performance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of Atmospheric Pressure • Factors affecting Atmospheric Temperature • Standard Atmosphere Model • Measurement of Atmospheric Pressure	L & Ds	PPT, IB	15 min
(b)	• Pressure variations in different layers • Temperature lapse rates • ISA deviations • Altimeter settings and pressure units	L & Ds	PPT, PDF	20 min
(c)	• Relationship between pressure and temperature • Impact on aircraft systems • Standard sea level conditions • Practical application of ISA	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the relationship between atmospheric pressure and altitude.
(b) Describe the standard temperature lapse rate in the troposphere.
(c) How is atmospheric pressure measured in aviation contexts?

12. Home Assignment and Follow Up	Prepare a brief note on how deviations from the Standard Atmosphere affect altimeter readings, referencing data from LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Servo altimeter and cabin altimeter.	Refer to the master syllabus reference list	LP 09

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min